

StormTracker v3.50 — Feature Reference

Weather Dashboard

- **Multi-source blending:** Combines Open-Meteo (GFS + HRRR), NWS API, and AWC METAR data, prioritizing the most accurate local source
- **Hero card:** Current temperature with "feels like", humidity, cloud cover, pressure (with trend arrow), precipitation, dew point, UV index, and freezing level
- **72-hour hourly forecast** and **7-day daily forecast** with NWS detailed text descriptions
- **48-hour trend charts:** Temperature, pressure, wind, and visibility plotted over time
- **Cloud base estimation:** Calculated from temperature–dewpoint spread (spread × 400 ft AGL)
- **Fog risk assessment:** Analyzes spread, wind speed, time of day, and cloud cover to rate radiation/advection fog probability
- **Atmospheric stability:** Rates conditions from Stable !' Conditionally Unstable !' Unstable based on temperature, humidity, and spread
- **Surface inversion detection:** Flags trapped pollutant / fog conditions from calm, clear overnight setups
- **Barometric pressure trend:** Shows rising/falling/steady with arrow indicators

Radar Map

- **Dual-source radar:** Toggle between NEXRAD (US high-res) and RainViewer (global coverage)
- **Radar animation:** Playback controls for past and forecast radar loops with adjustable speed
- **Storm zone polygons:** Color-coded precipitation areas rendered as map overlays
- **ILS approach cone:** Animated inbound storm path visualization
- **NHC tropical tracks:** Real-time hurricane and cyclone paths from the National Hurricane Center
- **NWS alert polygons:** Geographic bounds of active weather warnings drawn on the map
- **mPING ground-truth reports:** Citizen-reported real-time weather observations (NOAA/NSSL)
- **HD Scan:** 15-mile high-resolution localized radar analysis at zoom 12
- **Scan Here:** Relocate scan center to current map view center
- **Map controls:** Zoom, recenter, toggle radar source, distance unit toggle (MI/KM)

Storm Detection & Tracking

- **Automated radar scanning:** Periodic NEXRAD or RainViewer tile analysis to detect precipitation cells
- **Storm cell cards:** Each detected cell shows intensity (dBZ), distance, bearing, speed, and direction
- **Impact score (%):** Custom algorithm estimating threat level based on distance, bearing, and cell movement
- **ETA countdown:** Live-updating arrival timers for approaching storms
- **Cell tracking:** Compares consecutive scans to determine individual cell velocity
- **Terrain interaction:** Analyzes local elevation (valleys/ridges) to predict channeling or blocking effects
- **2.5D/3D storm view:** Isometric projection with heights corresponding to radar reflectivity
- **Storm alerts:** Browser notifications for approaching cells with configurable thresholds

Radar Sonar

- **Circular mini-map:** Radial sweep display showing storms as blips relative to user position
- **Configurable settings:** Sweep speed, fade duration, always-on mode, dot opacity, glow intensity, grid brightness, dBZ floor

- **Dot size by dBZ class:** Adjustable scaling for Light, Moderate, Heavy, Intense, and Extreme returns
- **Overlays:** Storm arrows, aloft wind indicator, lightning markers ("e 48 dBZ)
- **Zoom levels:** 15 / 20 / 30 / 40 / 50 / 60 / 70 / 80 mile radius

Wind Gauges

- **Five selectable gauge styles:**
- **Wind simulation:** Real-time animated wind speed/direction updates
- **Beaufort scale bar:** Color-coded force indicator (F0–F12) with label
- **Min/max tracking:** Session minimum and maximum wind speed display
- **Gust tracking:** Separate gust display with configurable gust window (30s / 60s / 2m / 5m)
- **Trend arrow:** Rising/falling/steady wind speed indicator

Station / METAR (Aviation Weather)

- **Digital weather console:** Dedicated tab mimicking a professional weather station
- **Raw METAR decode:** Parses and displays decoded observation data
- **Flight category badge:** VFR / MVFR / IFR / LIFR with color coding and limiting factor
- **Cloud base card:** Shows estimated cloud base (spread × 400 ft) with comparison arrow vs. METAR-reported ceiling — green !' if reported "e estimated, red !' if reported < estimated
- **Aviation metrics:** Density altitude, pressure altitude, altimeter setting
- **Tappable unit cycling:** Tap any value to cycle through all supported units
- **Nearby station list:** Shows closest METAR stations with distance

Alerts & Notifications

- **NWS weather alerts:** Active warnings, watches, and advisories for current location (US only)
- **SPC storm reports:** Tornado, hail, and wind reports from the Storm Prediction Center
- **Threat ticker:** Scrolling header bar that changes color (green !' yellow !' orange !' red) based on immediate local threat level
- **Configurable ticker speed:** Adjustable from 50% to 200% in settings
- **Browser push notifications:** Alerts for NWS warnings and storm cell approaches (works in background)
- **Custom threshold alerts:** User-defined triggers for wind speed, temperature, or storm intensity
- **Hazards section:** Local NWS hazard outlook

AI Weather Assistant

- **Chat interface:** AI-powered meteorological briefing using all current app data (METAR, storm ETAs, alerts, AFD)
- **Customizable persona:** Set tone to Professional, Friendly, or Humorous
- **Detail level control:** Adjustable verbosity for AI responses
- **Area Forecast Discussion (AFD):** Parses technical NWS forecaster discussions

Location & Navigation

- **GPS location:** Browser geolocation with accuracy display and altitude capture
- **Location search:** Smart address picker with autocomplete (Nominatim !' Photon !' Open-Meteo fallback)
- **Map pick:** Full-screen map with crosshair for precise location selection
- **Favorites:** Save up to 5 locations with rename support
- **Home location:** One-tap return to saved home position

- **Travel mode:** Live GPS tracking that auto-refreshes weather and radar as the user moves
- **Auto unit detection:** Switches between Imperial (US) and Metric based on detected country

Settings & Customization

- **Unit system:** Auto / Imperial / Metric / Custom presets with individual unit overrides for temperature (°F/°C), wind (mph/kts/km/h/m/s), pressure (inHg/mb/mmHg/kPa), visibility (mi/km), and precipitation (in/mm/cm)
- **Time format:** Auto / 12-hour / 24-hour
- **Gauge style selector:** Choose from 5 wind gauge designs
- **Wind simulation controls:** Adjustable update interval (5–30s), gust window, averaging window
- **Sonar settings:** Full configuration panel (see Radar Sonar section)
- **Neon weather icons:** Animated SVG/video weather icons
- **Scan radius:** Configurable storm detection radius
- **Radar metric toggle:** MI/KM distance display on radar
- **Desktop layout:** Responsive sidebar navigation, scaled-up text and controls on screens "e 1024px

Platform & Infrastructure

- **Static site on GitHub Pages:** All files served from `docs/` directory
- **Cloudflare Worker API:** Backend proxy at `stormtracker.joshua-622.workers.dev`
- **Service worker:** Offline caching with versioned cache (`stormtracker-v350`)
- **Multi-language support:** UI translation for 30+ languages via integrated translation engine
- **Progressive Web App:** Installable with manifest and icons
- **No server-side storage:** All user preferences stored in localStorage